

Introduction to Relational Event Models

Melania **Lembo**¹ Martina **Boschi**¹ Ernst C. **Wit**¹

¹Università della Svizzera italiana

🌐: <https://github.com/martinaboschi/rem-tutorial>

✉: melania.lembo@usi.ch - martina.boschi@usi.ch

Winter School in
Network Data Science and Artificial Intelligence
February 9, 2026



Università
della
Svizzera
italiana

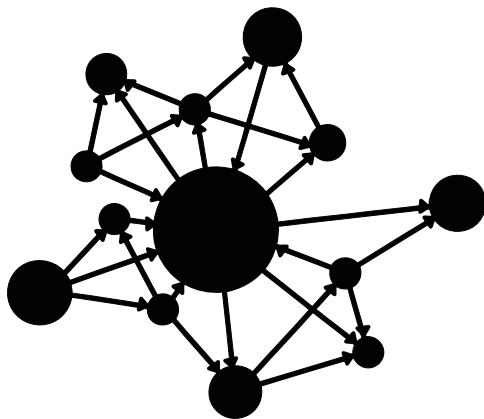
Agenda



Section	Presenter	Time	Abstract
Core REMs	Melania	13:00–13:40	Introduction to REM formulation and inclusion of time-aware – exogenous and endogenous – variables.
REM Inference	Martina	13:45–14:30	Overview of likelihood-based estimation methods.
Non-Linear REMs	Martina	14:35–15:20	Modelling non-linear and time-varying covariate effects while accounting for network and actor heterogeneity.
Practical	Melania	15:25–16:00	Applying REMs to empirical datasets using R Studio.



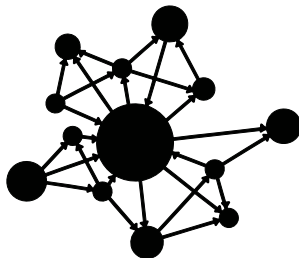
What are some phenomena you would naturally describe as a network?





What are some phenomena you would naturally describe as a network?

- ▶ email communications
- ▶ financial transactions
- ▶ non-native species invasions
- ▶ inter-hospital patient referral



Any of them *static*, for real?



What is a “longitudinal network”?

Longitudinal network: a network where edges have a temporal component:

- ▶ **relational state:** edges appear and persist in time (e.g. friendship)
- ▶ **relational event:** edges are “instantaneous” (e.g. communication)

In this workshop we will focus¹ on

relational event models

¹However relational states can be thought of as two relational events:

- ▶ **friendship start:** a relational event marking beginning of state
- ▶ **friendship end:** a relational event marking end of state



Modeling longitudinal networks

Let $\{N_{sr}(t)\}_{s \in V^S, r \in V^R, t \geq 0}$ be **longitudinal network**:

- ▶ $N_{sr}(t)$ = number of interactions from s to r until time t .
- ▶ $N_{sr}(0) = 0$

Mathematically, it is always possible to decompose:

$$N_{sr}(t) = \underbrace{\Lambda_{sr}(t)}_{\text{structural part}} + \underbrace{M_{sr}(t)}_{\text{noise part}}$$

If it exists, then the **hazard**

$$\lambda_{sr}(t) = \frac{d\Lambda_{sr}}{dt}(t)$$

is *instantaneous rate* that (s, r) occurs at time t .



What is a hazard?

Hazard (also: rate, intensity): “instantaneous probability” that event occurs now.

Mathematically:

$$P((s, r) \text{ occurs } (t, t + \delta t] \mid \text{did not occur until } t) \approx \lambda_{sr}(t)\delta t$$

So, if we know **hazard**:

- ▶ we know **entire probabilistic structure** of event network
- ▶ we can **simulate** relational events:
 - **Gillespie algorithm**: using exponential event times $\text{Exp}(\lambda_{sr})$.
 - **tau-leap algorithm**: using small time increments δt

Modeling hazard = modelling network



Let

$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{Is } (s, r) \text{ at risk?}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp\{f(x_{sr}(t), t)\}}_{\text{edge-specific risk factors}}$$

where

- ▶ $W_{sr}(t)$: edge-specific risk indicator
e.g. a past paper *cannot* cite a future paper.
- ▶ $\lambda_0(t)$: global risk determinants
e.g. very few emails sent when it's night at time t .
- ▶ $f(x_{sr}(t), t)$: edge-specific risk determinants
e.g. if r is a popular person, then (s, r) more likely.



How does a relational event network unfold?

Let's consider 5 individuals applying for a job:

- ▶ Each has “work experience”-score WE given by

0, 3, 5, 9, 11 years

- ▶ They need to get **two references letters** from each other.
- ▶ Person s contacts person r for a reference letter with rate,

$$\lambda_{sr}(t) = W_{sr}(t)e^{-1.5|WE_s - WE_r|}.$$

- ▶ Value -1.5 identifies
 - **relevance of WE:** it is not zero
 - **direction of WE:** larger WE difference $\Rightarrow$ fewer requests
- ▶ W_{sr} has value
 - 1 until person s has reached out to 2 people,
 - after which it returns to 0.



Distribution of event times



There are $5 \times 4 = 20$ requests “waiting” to be asked. Let

ΔT_{sr} = waiting time until s asks r for a letter

The distribution of ΔT_{sr} is given as

$$\Delta T_{sr} \sim \text{Exp}(e^{-1.5|WE_s - WE_r|})$$



When will first request be sent out?

The first request will happen at time T_1 , where

$$T_1 = \min_{sr} \Delta T_{sr}$$

Using property that **minimum of exponentials is exponential**,

$$T_1 \sim \text{Exp} \left(\sum_{s,r} e^{-1.5|WE_s - WE_r|} \approx 0.23 \right)$$





Which pair of individuals make up first request?

We now know that first event happened at t_1 . But which one?

E_1 = event that happened at time t_1 .

Using property that **which exponential is minimum is multinomial**,

$$E_1 \sim \text{Multinomial} \left(\frac{e^{-1.5|WE_1 - WE_2|}}{\sum_{s,r} e^{-1.5|WE_s - WE_r|}}, \dots, \frac{e^{-1.5|WE_5 - WE_4|}}{\sum_{s,r} e^{-1.5|WE_s - WE_r|}} \right)$$





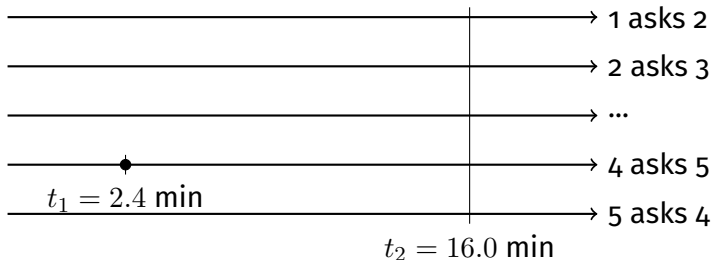
When will second request be sent out?

The second request will happen at time T_2 , where

$$T_2 = \min_{sr \neq s_1 r_1} \Delta T_{sr}$$

Using **memoryless property of an exponential**,

$$T_2 \sim T_1 + \text{Exp} \left(\sum_{(s,r) \neq (s_1, r_1)} e^{-1.5|WE_s - WE_r|} \approx 0.18 \right)$$





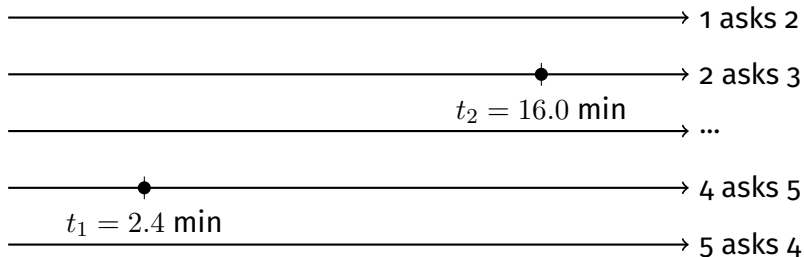
Which pair of individuals make up second request?

We know second event happened at $t_2 = 16.0$ min. Which one?

E_2 = event that happened at time t_2 .

Using property that **which exponential is minimum is multinomial**,

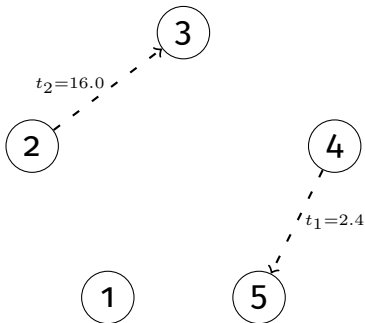
$$E_2 \sim \text{Multinomial} \left(\frac{e^{-1.5|WE_1 - WE_2|}}{0.18}, \dots, \frac{e^{-1.5|WE_5 - WE_4|}}{0.18} \right)$$



Repeat... to obtain a dynamic network



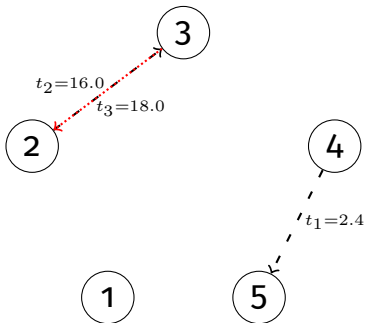
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



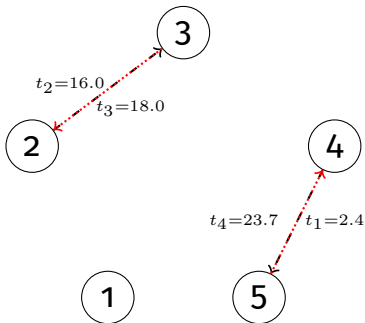
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



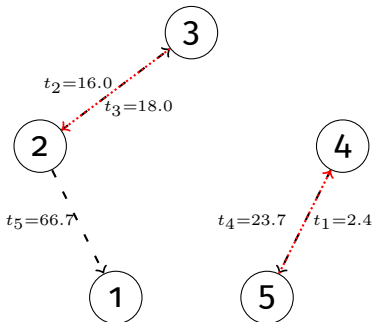
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



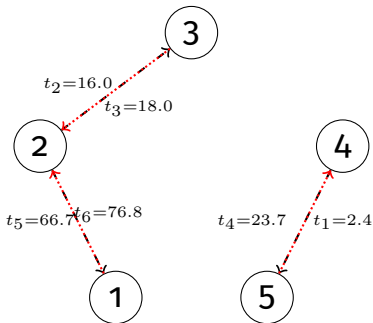
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



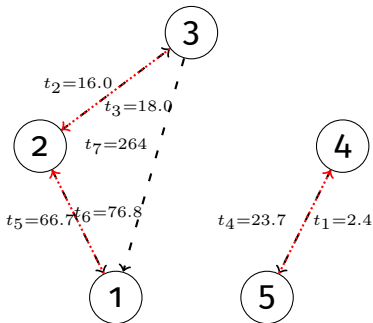
Repeating above procedure another 8 times, results in:





Repeat... to obtain a dynamic network

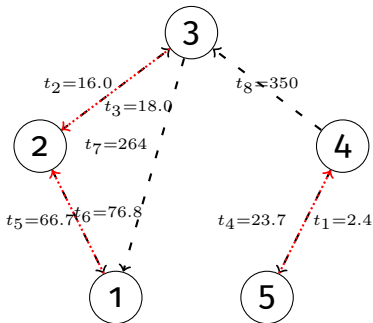
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



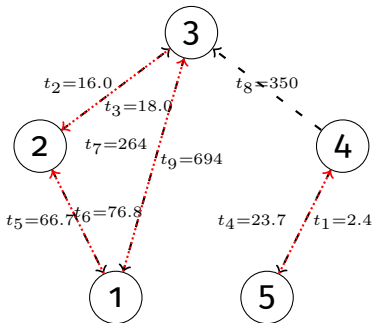
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



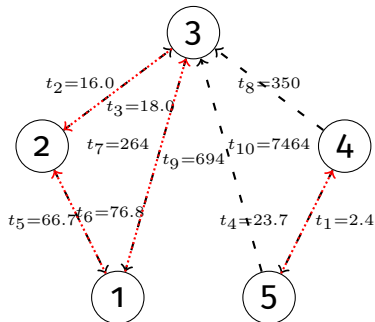
Repeating above procedure another 8 times, results in:



Repeat... to obtain a dynamic network



Repeating above procedure another 8 times, results in:





From Data to Answers

Let's assume that someone gave us this dynamic network:

	time	solicitor	receiver	WE_s	WE_r	$WE_{abs-diff}$
1	2.4	4	5	9	11	2
2	16.0	2	3	3	5	2
3	18.0	3	2	5	3	2
4	23.7	5	4	11	9	2
5	66.7	2	1	3	0	3
6	76.8	1	2	0	3	3
7	264.4	3	1	5	0	5
8	349.8	4	3	9	5	4
9	694.1	1	3	0	5	5
10	7463.6	5	3	11	5	6

Could we determine influence of WE on this dynamic? Yes, by doing **inference**!



Inference on what?

$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{Is } (s, r) \text{ at risk?}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \underbrace{\exp\{f(x_{sr}(t), t)\}}_{\text{edge-specific risk factors}}$$

We now focus on what the $x_{sr}(t)$ can be...

Also called:

- ▶ event statistics
- ▶ explanatory variables
- ▶ covariates
- ▶ risk factors
- ▶ drivers



Exogenous VS Endogenous drivers

Back to the example...

Work experience similarity $x_{sr}(t) = |WE_s - WE_r|$ depends on:

- ▶ Work experience score WE_s which is a node-level attribute, that does not depend on prior interactions → **exogenous** covariate
- ▶ Work experience similarity between s and r is dyadic, still exogenous.



Exogenous VS Endogenous drivers

Back to the example...

Work experience similarity $x_{sr}(t) = |WE_s - WE_r|$ depends on:

- ▶ Work experience score WE_s which is a node-level attribute, that does not depend on prior interactions → **exogenous** covariate
- ▶ Work experience similarity between s and r is dyadic, still exogenous.
- ▶ Age, gender, occupation, time-zone, distance and any function of them...

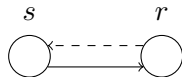


Exogenous VS Endogenous drivers

But prior events influence future events...

In the reference letters example: *I ask you, you ask me.*

$\rightsquigarrow$ Reciprocity (*binary*) $x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$



	time	solicitor	receiver	WE _{abs-diff}	REC _{binary}
1	2.4	4	5	2	0
2	16.0	2	3	2	0
3	18.0	3	2	2	1
4	23.7	5	4	2	1
5	66.7	2	1	3	0
6	76.8	1	2	3	1
7	264.4	3	1	5	0
8	349.8	4	3	4	0
9	694.1	1	3	5	1
10	7463.6	5	3	6	0

Exogenous VS Endogenous drivers



But prior events influence future events...

In the reference letters example: *I ask you, you ask me.*

$\rightsquigarrow$ Reciprocity (*binary*) $x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$



Depends on prior interactions $\rightarrow$ **endogenous** covariate

Exogenous VS Endogenous drivers



But prior events influence future events...

In the reference letters example: *I ask you, you ask me.*

$\rightsquigarrow$ Reciprocity (*binary*) $x_{sr}(t) = \mathbf{1}(\{(r, s, t^*) \mid t^* < t\} \neq \emptyset)$



Depends on prior interactions $\rightarrow$ **endogenous** covariate

Alternative choice for reciprocity definition:

- ▶ $x_{sr}(t) = \text{number of occurrences of } (r, s) \text{ before } t$
- ▶ $z_{sr}(t) = \begin{cases} t - \tau_{sr}, & \tau_{sr} = \text{time of last event } (r, s) \\ \infty, & \text{otherwise} \end{cases} \rightarrow \text{defined as **internal time**}$
- we transform to $x_{sr}(t) = \exp(-z_{sr}(t)) \rightsquigarrow$ the closer to zero the further in time, closer to one the opposite.

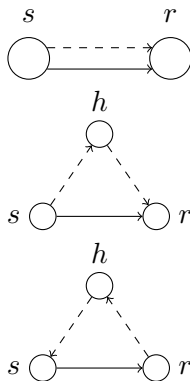
Exogenous VS Endogenous drivers



But prior events influence future events...

Similar kind of definitions can be used for various endogenous mechanisms:

- ▶ Repetition (*I ask you, I ask you again*)
- ▶ Triadic closure (*I ask you, you ask them, so I ask them*)
- ▶ Cyclic closure (*They ask you, you ask me, so I ask them*)
- ▶ ...

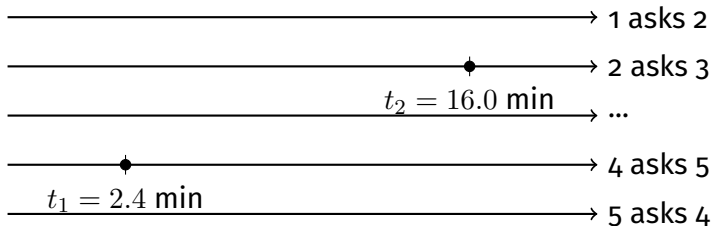




Versatility isn't cheap...

Endogenous variables powerful for dependence between edges but computationally intensive to compute:

- ▶ n relational events and $|V^S| = |V^R| = p \rightsquigarrow p^2$ possible interactions
- ▶ for each edge, look at all previous events and check which satisfy the condition of interest... on average $\frac{n}{2}$
- ▶ $\implies$ Complexity is $O(n^2 p^2)$ in the worst case, just for dyadic...





Inference of what?

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{f(x_{sr}(t), t)\}$$

Consider $f(x_{sr}(t), t) = \beta \cdot x_{sr}(t)$:

- ▶ Work experience similarity $x_{sr}(t) = |WE_s - WE_r| \rightsquigarrow \beta < 0$ the greater the difference in work experience, the less likely it is to ask for reference letter
- ▶ Reciprocity $x_{sr}(t) = \exp(-(t - \tau_{sr})) \rightsquigarrow \beta > 0$ the more recent the email, the more likely it's replied to soon (in principle...)

Estimating effect β given data = inference

Descending into Dante's Inferno...



Abandon hope all ye who enter here...

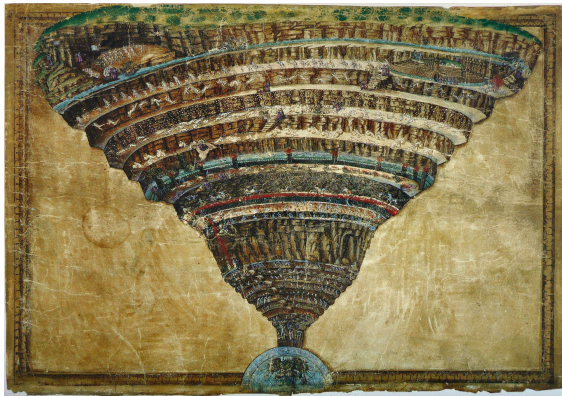
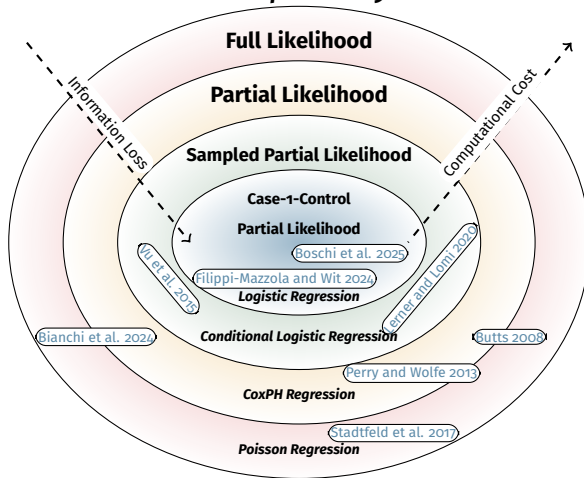


Fig. 1: Watts, Barbara J.: “Sandro Botticelli’s Drawings for Dante’s Inferno”



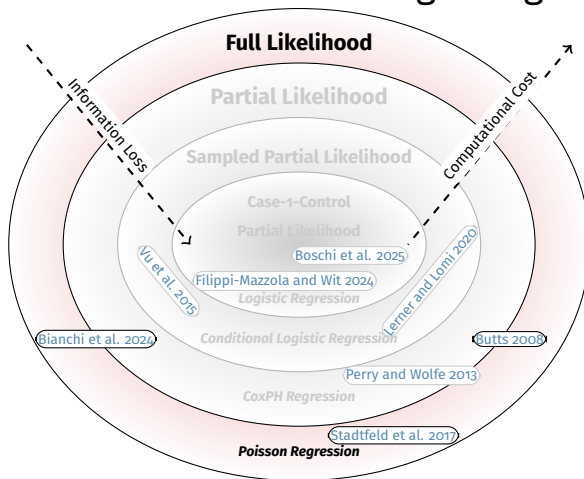
...or through rings of Likelihood?

Maintain Abandon hope all ye who enter here...





Let's start from the beginning...



A quick recap!



Relational Data: stream of observed events.

$$E = \{(s_i, r_i, t_i), i = 1, \dots, n\}$$

Parametric Model: linear (in log-hazard)... just for now!

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{\beta \cdot \mathbf{x}_{sr}(t)\}$$

Likelihood: constructed as joint probability of observing **data** under **model**.

AIM: estimate β that maximizes **likelihood**.



Full Likelihood: Let's follow the definition...

Likelihood: constructed as joint probability of observing **data** under **model**.

Perry and Wolfe [2013]

$$\mathcal{L}(\beta) = p(E; \beta) \quad \underbrace{\quad}_{\text{Multiplication Theorem}} \quad \prod_{i=1}^n \left[p\left(t_i \mid \underbrace{t_1, (s_1, r_1), \dots, t_{i-1}, (s_{i-1}, r_{i-1})}_{\text{previous events}}; \beta\right) \times \right. \\ \left. p\left(\underbrace{(s_i, r_i) \mid t_i, t_1, (s_1, r_1), \dots, t_{i-1}, (s_{i-1}, r_{i-1})}_{\text{previous events}}; \beta\right) \right]$$

events are independent **conditional on history**
along with covariates, incorporated by $\mathbb{W} = \{\mathcal{W}_t\}_{t \in \mathbb{R}^+}$

$$= \prod_{i=1}^n p(t_i \mid \mathcal{W}_{t_i^-}; \beta) p((s_i, r_i) \mid t_i, \mathcal{W}_{t_i^-}; \beta)$$



$$\mathcal{L}(\beta) = \prod_{i=1}^n p(t_i | \mathcal{W}_{t_i^-}) \times p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \beta)$$

How to account for fact that there are not other dyads observed in (t_{i-1}, t_i) ?

$$\underbrace{\prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) du \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \times \underbrace{p((s_i, r_i) | t_i, \mathcal{W}_{t_i^-}; \beta)}_{\text{Multinomial Probability}} = \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$

Full Likelihood for... Piecewise Constant Rates (PCR) I



$$\mathcal{L}(\beta) = \underbrace{\prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) du \right]}_{\text{Minimum Generalized Exponential Inter-arrival Time}} \times \underbrace{\frac{p((s_i, r_i) | t_i, \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}}_{\text{Multinomial Probability}}$$

What if **hazard is assumed to be constant within inter-arrival times?**

$$\prod_{i=1}^n \underbrace{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right]}_{\text{Exponential waiting time } \Delta T_i = T_i - T_{i-1} \text{ - Poisson N. of Events in } \Delta T_i} \times \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$

Full Likelihood for... Piecewise Constant Rates (PCR) II



$$\begin{aligned}
 \mathcal{L}(\beta) &\stackrel{\text{PCR}}{=} \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right] \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)} \\
 &\quad \prod_{i=1}^n \lambda_{s_i r_i}(t_i; \beta) \prod_{(s,r) \in \mathcal{R}_{t_i}} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})]
 \end{aligned}$$

Full Likelihood for... Piecewise Constant Rates (PCR) III



$$\begin{aligned}
 \mathcal{L}(\beta) &\stackrel{\text{PCR}}{=} \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \times \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) (t_i - t_{i-1}) \right] \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)} \\
 &\quad \prod_{i=1}^n \lambda_{s_i r_i}(t_i; \beta) \prod_{(s,r) \in \mathcal{R}_{t_i}} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})] \\
 &\quad \text{Call } \Delta N_{sr}(t_i) = N_{sr}(t_i) - N_{sr}(t_{i-1}) \\
 &\quad \prod_{i=1}^n \prod_{(s,r) \in \mathcal{R}_{t_i}} (\lambda_{sr}(t_i; \beta))^{\Delta N_{sr}(t_i)} \exp [-\lambda_{sr}(t_i; \beta) (t_i - t_{i-1})] \\
 &\quad \underbrace{\hspace{15em}}_{\text{Proportional to likelihood of a Poisson Regression}}
 \end{aligned}$$

So, which statistical model should we focus on? (Pt. 1)



Full Log-Likelihood (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{\beta \cdot \mathbf{x}_{sr}(t)\}$$

Inference:

Poisson Regression

$$\Delta N_{sr}(t_i) | \mathbf{x}_{sr}(t_i) \stackrel{\text{iid}}{\sim} \text{Poisson}(\mu_{sr}(t_i))$$

$$\log(\mu_{sr}(t_i)) = \beta \cdot \mathbf{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

So, which statistical model should we focus on? (Pt. 1)



Full Log-Likelihood (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp\{\beta \cdot \mathbf{x}_{sr}(t)\}$$

Inference:

Poisson Regression

$$\Delta N_{sr}(t_i) | \mathbf{x}_{sr}(t_i) \stackrel{\text{iid}}{\sim} \text{Poisson}(\mu_{sr}(t_i))$$

$$\log(\mu_{sr}(t_i)) = \beta \cdot \mathbf{x}_{sr}(t_i) + \log(t_i - t_{i-1})$$

Nice... But at what **price**?

1. Strong Assumption;
2. High Computational Cost;



Full Likelihood in Practice I

Full Log-Likelihood (Piecewise Constant Rates):

$$\ell(\beta) = \sum_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \Delta N_{sr}(t_i) \log(\lambda_{sr}(t_i; \beta)) - \lambda_{sr}(t_i; \beta)(t_i - t_{i-1})$$

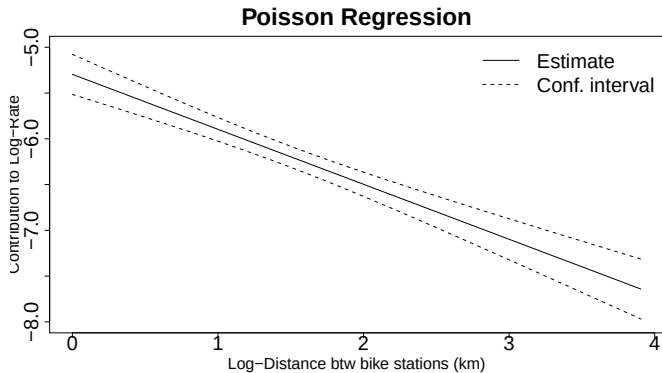
```
> head(poisson_dat, 10)
  sender receiver dist_min  dist  PT_dist random_intercept_s info event  start      stop log_dist  logtm
1  33202   31261   13.3  2766  3.2714157        3.46    80    1  0.0074  0.007485937  1.326013 -9.361895
2  31937   31937    0.0    0  1.7116404       -0.14     1    0  0.0074  0.007485937  0.000000 -9.361895
3  33200   31937   19.0  5258  4.1927036        0.85     2    0  0.0074  0.007485937  1.833861 -9.361895
4  32604   31937   38.9  9260  6.2539595       -7.03     3    0  0.0074  0.007485937  2.328253 -9.361895
5  31261   31937   17.6  4158  1.4511050        1.06     4    0  0.0074  0.007485937  1.640549 -9.361895
6  31246   31937   11.3  3333  1.4195458       -2.21     5    0  0.0074  0.007485937  1.466260 -9.361895
7  31285   31937   15.9  4465  2.3303198        0.46     6    0  0.0074  0.007485937  1.698364 -9.361895
8  31297   31937   13.7  3982  0.7931991        1.40     7    0  0.0074  0.007485937  1.605831 -9.361895
9  31218   31937   27.2  6001  1.9189675       -4.05     8    0  0.0074  0.007485937  1.946053 -9.361895
10 31966   31937   45.9 11088  1.8293264       -1.11     9    0  0.0074  0.007485937  2.492213 -9.361895
```



```
glm(event ~ log_dist + offset(logtm),
family = poisson(link = "log"), data = poisson_dat)
```



Full Likelihood in Practice II





Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better?



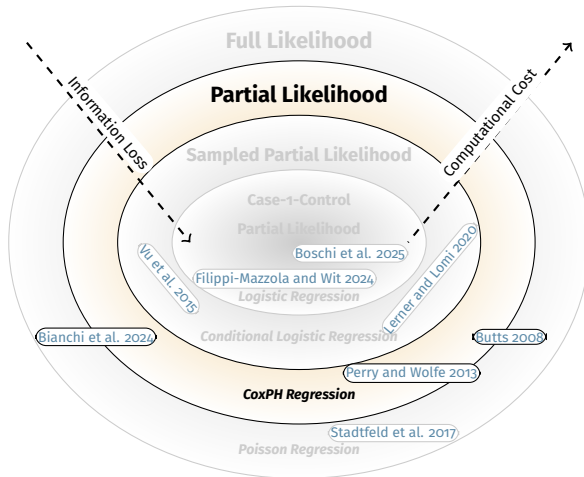
Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better? Let's do one step back:

$$\mathcal{L}(\beta) = \prod_{i=1}^n \sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta) \exp \left[- \sum_{(s,r) \in \mathcal{R}_{t_i}} \int_{t_{i-1}}^{t_i} \lambda_{sr}(u; \beta) du \right] \times \frac{\lambda_{s_i r_i}(t_i; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \beta)}$$

Partial Likelihood Inference





Inspired by Cox [1975]: REM's Partial Likelihood I

Partial likelihood focuses on **multinomial probability of observing a specific dyad among those at risk, given that an event has occurred**.

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$



Inspired by Cox [1975]: REM's Partial Likelihood II

Partial likelihood focuses on **multinomial probability of observing a specific dyad among those at risk, given that an event has occurred**.

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i | \mathcal{W}_{t_i}^-; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i | \mathcal{W}_{t_i}^-; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$

Information Loss:

Leaving $\lambda_0(t)$ arbitrary
results in
slight loss of information
about β .

Dimensionality reduction:

effective when providing a
simplification
of full likelihood by
eliminating
nuisance parameters.

Event Ordering:

Applicable when
only order of
events is known,
not their exact timing.

So, which statistical model should we focus on? (Pt. 2)



Partial Log-likelihood:

$$\ell^P(\beta) = \log \mathcal{L}^P(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{ \beta \cdot \mathbf{x}_{sr}(t) \}$$

Inference:

CoxPH Regression

$$\hat{\beta} = \arg \max_{\beta} \ell^P(\beta)$$



Partial Likelihood in Practice I

Partial Log-Likelihood:

$$\ell^P(\beta) = \log \mathcal{L}^P(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

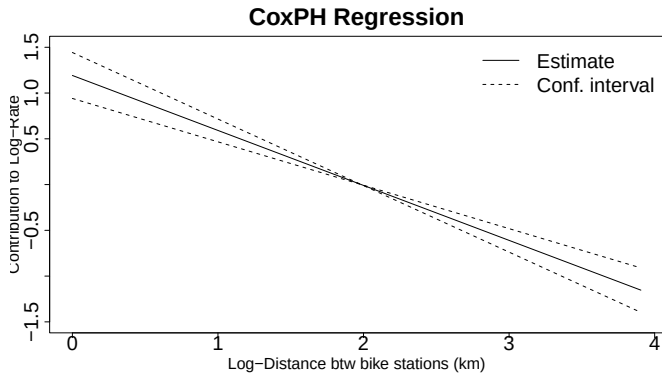
```
> head(cox_dat, 10)
  sender receiver dist_min  dist  PT_dist random_intercept_s info event  start      stop log_dist
1  33202   31261   13.3  2766  3.2714157         3.46    80    1 0.0074 0.007485937 1.326013
2  31937   31937    0.0    0  1.7116404        -0.14    1    0 0.0074 0.007485937 0.000000
3  33200   31937   19.0  5258  4.1927036         0.85    2    0 0.0074 0.007485937 1.833861
4  32604   31937   38.9  9260  6.2539595        -7.03    3    0 0.0074 0.007485937 2.328253
5  31261   31937   17.6  4158  1.4511050         1.06    4    0 0.0074 0.007485937 1.640549
6  31246   31937   11.3  3333  1.4195458        -2.21    5    0 0.0074 0.007485937 1.466260
7  31285   31937   15.9  4465  2.3303198         0.46    6    0 0.0074 0.007485937 1.698364
8  31297   31937   13.7  3982  0.7931991         1.40    7    0 0.0074 0.007485937 1.605831
9  31218   31937   27.2  6001  1.9189675        -4.05    8    0 0.0074 0.007485937 1.946053
10 31966   31937   45.9 11088  1.8293264        -1.11    9    0 0.0074 0.007485937 2.492213
```



```
coxph(Surv(start,stop,event) ~ log_dist, data = cox_dat)
```



Partial Likelihood in Practice II





Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better?



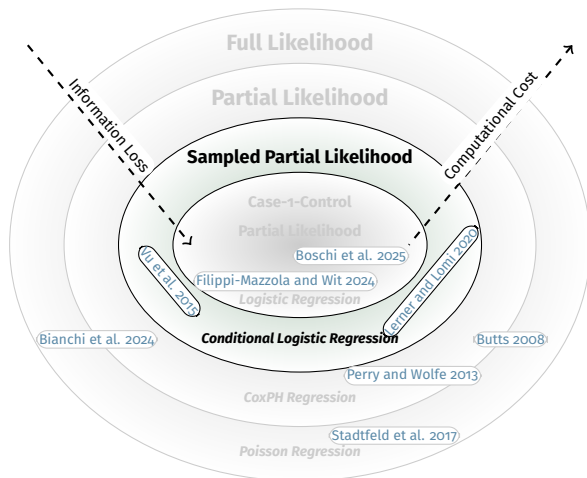
Full Likelihood via Poisson Regression... At what price?

1. Strong Assumption;
2. High Computational Cost;

Can we do better? Let's do one step back:

$$\mathcal{L}^P(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i; \mathcal{W}_{t_i^-}; \beta)}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \lambda_{sr}(t_i; \mathcal{W}_{t_i^-}; \beta)} = \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\sum_{(s,r) \in \mathcal{R}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \}}$$

Sampled Partial Likelihood Inference





What changes when we sample? I

Relational Data: stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$ is **sampled risk set** at time t_i .



What changes when we sample? I

Relational Data: stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$ is **sampled risk set** at time t_i .

Updated Filtration: incorporating history of events and sampling information.

$$\{\mathcal{F}_t\}_{t \geq 0}, \quad \mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \leq t)$$



What changes when we sample? I

Relational Data: stream of observed events and sampled risk sets.

$$D_m = \{(s_i, r_i, t_i, \mathcal{SR}_{t_i}), |\mathcal{SR}_{t_i}| = m + 1, i = 1, \dots, n\},$$

where $\mathcal{SR}_{t_i} \subset \mathcal{R}_{t_i}$ is **sampled risk set** at time t_i .

Updated Filtration: incorporating history of events and sampling information.

$$\{\mathcal{F}_t\}_{t \geq 0}, \quad \mathcal{F}_t = \mathcal{W}_t \cup \sigma(\mathcal{SR}_{t_i}; t_i \leq t)$$

Parametric Model: let's stay linear for the moment...

$$\lambda_{sr, \mathcal{SR}}(t) = \lambda_{sr}(t) \cdot \pi_t(\mathcal{SR} | (s, r)), \quad \pi_t(\mathcal{SR} | (s, r)) = \frac{m}{|\mathcal{R}_t| - 1} \cdot 1_{\{(s, r) \in \mathcal{SR}, \mathcal{SR} \subset \mathcal{R}_t\}}$$

m is the **number of sampled non-events** according to **nested case-control sampling**.



What changes when we sample? II

Sampled Partial likelihood:

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\lambda_{s_i r_i}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s_i, r_i))}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \lambda_{sr}(t_i; \mathcal{F}_{t_i}^-; \beta) \cdot \pi_{t_i}(\mathcal{SR}_{t_i} | (s, r))} = \prod_{i=1}^n \frac{\exp\{\beta \cdot x_{s_i r_i}(t_i)\}}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp\{\beta \cdot x_{sr}(t_i)\}}$$

Look Familiar?

Conditional logistic regression

[🌐 1 language](#)
[Article](#)
[Talk](#)
[Read](#)
[Edit](#)
[View history](#)
[Tools](#)

From Wikipedia, the free encyclopedia

Conditional logistic regression is an extension of [logistic regression](#) that allows one to account for [stratification](#) and [matching](#). Its main field of application is [observational studies](#) and in particular [epidemiology](#). It was devised in 1978 by [Norman Breslow](#), [Nicholas Day](#), [Katherine Halvorsen](#), [Ross L. Prentice](#) and [C. Sabat](#).^[1] It is the most flexible and general procedure for matched data.

Background [\[edit \]](#)

Observational studies use [stratification](#) or [matching](#) as a way to control for [confounding](#).

[Logistic regression](#) can account for stratification by having a different constant term for each stratum. Let us denote $Y_{it} \in \{0, 1\}$ the label (e.g. case status) of the t th observation of the i th stratum and $X_{it} \in \mathbb{R}^p$ the values of the corresponding predictors. We then take the likelihood of one observation to be

$$P(Y_{it} = 1 | X_{it}) = \frac{\exp(\alpha_i + \beta^\top X_{it})}{1 + \exp(\alpha_i + \beta^\top X_{it})}$$

Fig. 2: Source: Wikipedia

Stratification...
with respect to what?
By case!

So, which statistical model should we focus on? (Pt. 3)



Sampled Partial likelihood:

$$\mathcal{L}^S(\beta) = \prod_{i=1}^n \frac{\exp \{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp \{\beta \cdot \mathbf{x}_{sr}(t_i)\}}$$

Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{\beta \cdot \mathbf{x}_{sr}(t)\}$$

Inference:

Conditional Logistic Regression

$$\Delta N_{sr}(t_i) | \sum_{s',r' \in \mathcal{SR}_{t_i}} \Delta N_{s'r'}(t_i) = 1 \sim \text{Categorical}(\pi_{sr}(t_i))$$

$$\pi_{sr}(t_i) = \frac{\exp \{\beta \cdot \mathbf{x}_{sr}(t_i)\}}{\sum_{(s',r') \in \mathcal{SR}_{t_i}} \exp \{\beta \cdot \mathbf{x}_{s'r'}(t_i)\}}$$

Sampled Partial Likelihood in Practice I



Sampled Partial Log-Likelihood:

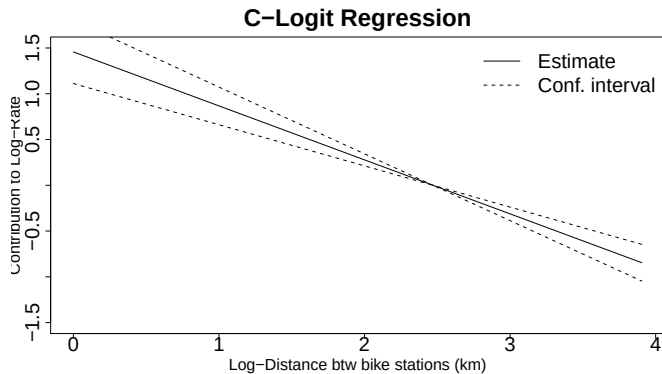
$$\ell^S(\beta) = \log \mathcal{L}^S(\beta) = \sum_{i=1}^n \left[\beta \cdot \mathbf{x}_{s_i r_i}(t_i) - \log \left(\sum_{(s,r) \in \mathcal{SR}_{t_i}} \exp \{ \beta \cdot \mathbf{x}_{sr}(t_i) \} \right) \right],$$

```
> head(sampled_dat, 10)
```

	sender	receiver	dist_min	dist	PT_dist	random_intercept_s	info	event	start	stop	log_dist
1	33202	31261	13.3	2766	3.2714157	3.46	80	1	0.0074	0.007485937	1.326013
2	31297	31218	15.9	3630	0.7931991	1.40	147	0	0.0074	0.007485937	1.532557
3	31255	31419	19.5	4453	6.7501025	-0.15	318	0	0.0074	0.007485937	1.696166
4	31071	32604	50.3	12994	1.6706971	3.96	54	0	0.0074	0.007485937	2.638629
5	31272	31071	34.6	8183	2.2255955	0.14	277	0	0.0074	0.007485937	2.217354
6	32217	31071	124.3	33190	3.8203727	-3.86	279	0	0.0074	0.007485937	3.531933
7	31324	31262	11.2	2435	1.3001624	6.14	250	1	0.0132	0.013290035	1.234017
8	32217	31297	117.9	44455	3.8203727	-3.86	139	0	0.0132	0.013290035	3.816723
9	31255	32604	49.4	12196	6.7501025	-0.15	58	0	0.0132	0.013290035	2.579914
10	32604	31218	63.9	15186	6.2539595	-7.03	143	0	0.0132	0.013290035	2.784147



```
clogit(event ~ log_dist + strata(stop), data = sampled_dat)
```



Who first used this approach in REMs?



Social Networks

Volume 43, October 2015, Pages 121-135



Relational event models for social learning in MOOCs

Duy Vu ^a  , Philippa Pattison ^b  , Garry Robins ^a  



Network Science

Reliability of relational event model estimates under sampling: How to fit a relational event model to 360 million dyadic events

Published online by Cambridge University Press: 22 November 2019

Jürgen Lerner  and Alessandro Lomi

Show author details 

Article Metrics



Towards Case-control sampling via GLM

Full Likelihood via Poisson Regression... At what price?

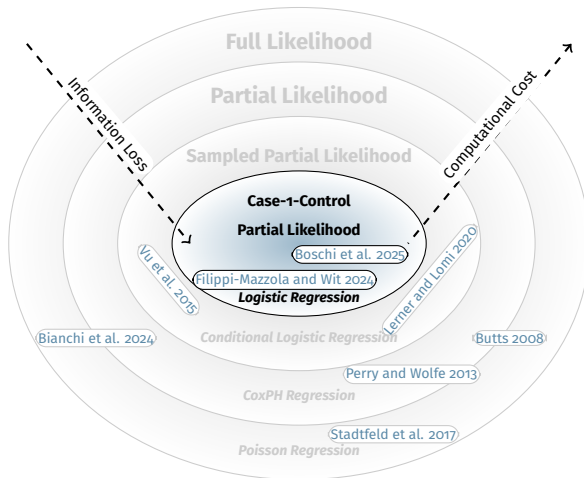
1. ~~Strong Assumption;~~
2. ~~High Computational Cost;~~

Well, we can do we even better!



Case-1-control Sampling Inference

Can we do more?





How to get a Logistic Regression... I

Sampled Partial likelihood with $m = 1$:

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \}}{\exp \{ \beta \cdot \mathbf{x}_{s_i r_i}(t_i) \} + \exp \{ \beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i) \}}$$



How to get a Logistic Regression... II

Sampled Partial likelihood with $m = 1$:

$$\begin{aligned}
 \mathcal{L}^S(\beta) &\stackrel{m=1}{=} \prod_{i=1}^n \frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\} + \exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} \\
 &= \prod_{i=1}^n \frac{\frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}}{\frac{\exp\{\beta \cdot \mathbf{x}_{s_i r_i}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}} + \frac{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}{\exp\{\beta \cdot \mathbf{x}_{s_i^* r_i^*}(t_i)\}}} = \prod_{i=1}^n \frac{\exp\{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}}{1 + \exp\{\beta \cdot (\mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i))\}} \\
 &= \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta \mathbf{x}_i) \quad \Delta \mathbf{x}_i = \mathbf{x}_{s_i r_i}(t_i) - \mathbf{x}_{s_i^* r_i^*}(t_i)
 \end{aligned}$$

So, which statistical model should we focus on? (Pt. 4)



Sampled Partial likelihood with $m = 1$:

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta x_i) \quad \Delta x_i = x_{s_i r_i}(t_i) - x_{s_i^* r_i^*}(t_i)$$

Modeling:

Relational Event Model

$$\lambda_{sr}(t) = W_{sr}(t) \times \lambda_0(t) \times \exp \{ \beta \cdot x_{sr}(t) \}$$

Inference:

Logistic Regression

$$Y_i | \Delta x_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi_i)$$

$$\text{logit}(\pi_i) = \beta \cdot \Delta x_i$$

$$y_i = 1 \quad i = 1, \dots, n; \quad \text{no intercept}$$

Wondering what advantage is? **Linearity** no more!

Case-1-control Sampling Inference via GLM in Practice I



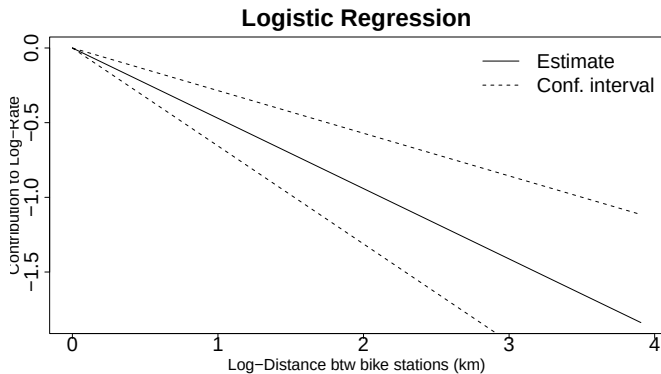
Sampled Partial likelihood with $m = 1$:

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \text{logistic}(\beta \cdot \Delta x_i) \quad \Delta x_i = x_{s_i r_i}(t_i) - x_{s_i^* r_i^*}(t_i)$$

```
> head(dat_gam, 10)
  y      time  s1  r1 log_dist1  s2  r2 log_dist2 delta_log_dist
1 1 0.007485937 33202 31261 1.3260134 33200 33202 1.229348 0.096665320
2 1 0.013290035 31324 31262 1.2340169 31966 31255 2.766130 -1.532113487
3 1 0.016123561 31261 33200 0.6333972 31272 31071 2.217354 -1.583956772
4 1 0.016833676 31218 31218 0.0000000 31218 32604 2.821557 -2.821557442
5 1 0.027780092 31285 31246 0.7788661 31297 31071 1.876101 -1.097234562
6 1 0.036779525 31272 31023 2.3787125 31966 32050 3.278728 -0.900015931
7 1 0.045849812 31218 31218 0.0000000 31272 31262 1.149305 -1.149305403
8 1 0.047972385 33202 31966 2.8733951 31255 31937 1.585350 1.288045017
9 1 0.052986957 31272 31419 1.9516082 33200 31313 1.925853 0.025754966
10 1 0.061285646 31071 31023 1.9742199 31071 31246 1.968091 0.006129007
```



```
gam(y ~ -1 + delta_log_dist,
family="binomial"(link = 'logit'), data = dat_gam)
```

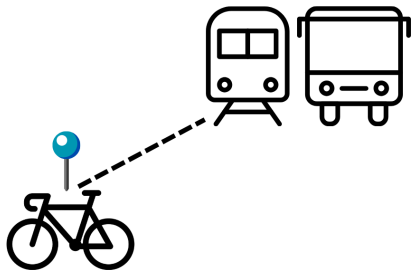


All you need is linear, right?



LET'S TALK BIKE SHARING DYNAMICS

How does distance between a bike station and a public transportation hub affect the number of bike rides? Is this effect going to be the same at 8AM or 6PM compared to 2 PM?



We need **time-varying effects**: the risk factor x_{sr} has **different effect across time**

$$\beta(t) \cdot x_{sr}(t)$$

All you need is linear, right?



LET'S TALK BIKE SHARING DYNAMICS

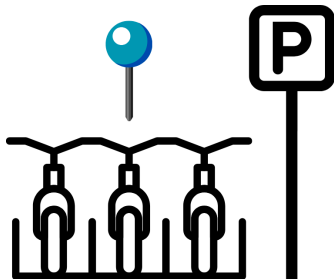
Do people bike far? What is the effect of biking 1km more if I have to bike 2kms compared to 20kms?

We need **non-linear effects**: **different effect for different values** assumed by the risk factor x_{sr}

$$f(x_{sr}(t))$$



All you need is linear, right?



LET'S TALK BIKE SHARING DYNAMICS

Not all bike stations are the same! Some have higher/lower usage due to aspects that are unrelated to the predictors we account for (e.g. bike availability, proximity to tourist attractions...).

We need **random effects**: accounts for the **heterogeneity** arising from the system features

$$\gamma \cdot v_{sr}(t)$$

Back to the hazard



$$\lambda_{sr}(t) = \underbrace{W_{sr}(t)}_{\text{risk indicator}} \times \underbrace{\lambda_0(t)}_{\text{baseline hazard}} \times \exp \left\{ \underbrace{f(x_{sr}(t), v_{sr}(t), t)}_{\text{contribution function}} \right\}$$

$$f(x_{sr}(t), v_{sr}(t), t) = \underbrace{\beta \cdot x_{sr}^{(1)}(t)}_{\text{fixed linear}} + \underbrace{\beta(t) \cdot x_{sr}^{(2)}(t)}_{\text{time-varying}} + \underbrace{f(x_{sr}^{(3)}(t))}_{\text{non-linear}} + \underbrace{\gamma \cdot v_{sr}(t)}_{\text{random}}$$

1. **fixed linear**: β is time-homogenous and is not a function of $x_{sr}^{(1)}(t)$;
2. **time-varying**: $\beta(t)$ is a non-linear function of time;
3. **non-linear**: $f(x_{sr}^{(3)}(t))$ is a non-linear function of $x_{sr}^{(3)}(t)$;
4. **random**: γ from a Gaussian random variable;



Reminder!

Fixed linear effects are fitted using a **logistic regression** with:

- ▶ only successes ($y_i = 1, i = 1, \dots, n$);
- ▶ no intercept;
- ▶ covariates: $\Delta x_i := x_{s_i} r_i - x_{s_i^*} r_i^*$;

using `glm(..., family = "binomial")` function.

We relax the assumption of linearity and exploit **generalized additive models** instead. Implemented in `gam` function of package `mgcv`.

mgcv: Mixed GAM Computation Vehicle with Automatic Smoothness Estimation

Generalized additive (mixed) models, some of their extensions and other generalized ridge regression with multiple smoothing parameter estimation by (Restricted) Marginal Likelihood, Generalized Cross Validation and similar, or using iterated nested Laplace approximation for fully Bayesian inference. See Wood (2017) <[doi:10.1201/9781315370279](https://doi.org/10.1201/9781315370279)> for an overview. Includes a `gam()` function, a wide variety of smoothers, 'JAGS' support and distributions beyond the exponential family.





Modeling Time-Varying Effects

- ▶ Effect of $x_{sr}^{(2)}(t)$ as $\rightarrow$ **Spline function of Time**

$$\beta(t) \approx \sum_{j=1}^q \theta_j b_j(t)$$

Choices to be made:

- ▶ **basis dimension** q
- ▶ a **smooth basis functions** $\{b_j\}_j$ e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate q **smooth non-linear functions** of time $b_j(t)$.

Estimating $\beta(t) \iff$ Estimating θ_j -s

TVE: contribution of distance from public transport

► ... to **log-hazard of an interaction**

$$f(x_{sr}(t), v_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(t)}_{\text{time-varying effect}} \times \underbrace{x_{sr}^{(2)}(t)}_{\text{covariate}} + \dots$$

► ... to **sampled partial likelihood (m=1)**

$$\mathcal{L}^S(\beta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \beta(t_i)[x_{s_i r_i}^{(2)}(t_i) - x_{s_i^* r_i^*}^{(2)}(t_i)] + \dots}}{1 + e^{\dots + \beta(t_i)[x_{s_i r_i}^{(2)}(t_i) - x_{s_i^* r_i^*}^{(2)}(t_i)] + \dots}}$$

$$\downarrow$$

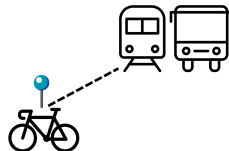
$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{ smoothing penalty})$$

TVE: contribution of distance from public transport

► ... to **sampled partial likelihood (m=1)**

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{ smoothing penalty})$$

```
> head(data)
  y    time    s1    r1 PT_dist1    s2    r2 PT_dist2    PT_dist
1 1 0.001451090 32217 32604 3.820373 32604 31937 6.2539595 -2.4335868
2 1 0.001584378 31937 31285 1.711640 32217 31246 3.8203727 -2.1087323
3 1 0.008190801 31324 31218 1.300162 31272 31261 2.2255955 -0.9254332
4 1 0.011731796 31313 31272 2.346278 31297 32604 0.7931991 1.5530784
5 1 0.014548734 31285 32217 2.330320 31261 31285 1.4511050 0.8792148
6 1 0.018378723 33200 31313 4.192704 31262 31419 1.1147747 3.0779289
```

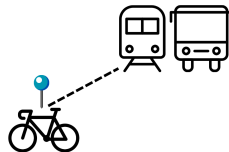
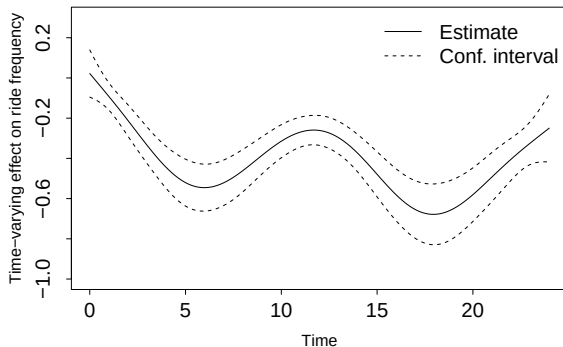


```
gam(y ~ -1+ s(time, by = PT_dist), family= "binomial", data = data)
```

TVE: contribution of distance from public transport

► ... to **sampled partial likelihood** ($m=1$)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}}{1 + e^{\dots + [\sum_{j=1}^q \theta_j b_j(t_i)] \Delta x_i^{(2)} + \dots}} \quad (+ \text{ smoothing penalty})$$



Modeling Non-Linear Effects



- ▶ Effect of $x_{sr}^{(3)}(t)$ as $\rightarrow$ **Spline function of Covariate**

$$f(x_{sr}^{(3)}(t)) \approx \sum_{j=1}^q \theta_j b_j(x_{sr}^{(3)}(t))$$

Choices to be made:

- ▶ **basis dimension** q
- ▶ a **set of smooth basis functions** $\{b_j\}_j$ e.g. *B-splines, thin plate regression splines,...*
- ▶ given range of variable, **basis evaluation points** (e.g. *uniform, quantile of a distribution,...*) to evaluate q **smooth non-linear functions** of covariate $b_j(x_{sr}^{(3)}(t))$.

Estimating $f \iff$ Estimating θ_j -s



NLE: contribution of distance btw two bike stations

- ... to **log-hazard of an interaction**

$$f(\mathbf{x}_{sr}(t), t) = \dots + \underbrace{\sum_{j=1}^q \theta_j b_j(x_{sr}^{(3)}(t))}_{\text{non-linear effect}} + \dots$$

- ... to **sampled likelihood function (m=1)**

$$\mathcal{L}^S(\boldsymbol{\theta}) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$

NLE: contribution of distance btw two bike stations



► ... to sampled likelihood function (m=1)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$

```
> head(data)
```

	y	time	s1	r1	dist1	s2	r2	dist2	ev.i	nonev.i
1	1	0.001377150	33202	32604	14.073	32604	31937	9.260	1	-1
2	1	0.001453772	31937	31285	4.404	32217	31246	43.790	1	-1
3	1	0.001979046	33200	31937	5.258	31255	31261	1.881	1	-1
4	1	0.004495738	31246	31937	3.333	31218	31937	6.001	1	-1
5	1	0.007457826	31324	31218	3.970	31272	31261	3.300	1	-1
6	1	0.010580056	31272	31324	4.492	31272	31937	6.689	1	-1



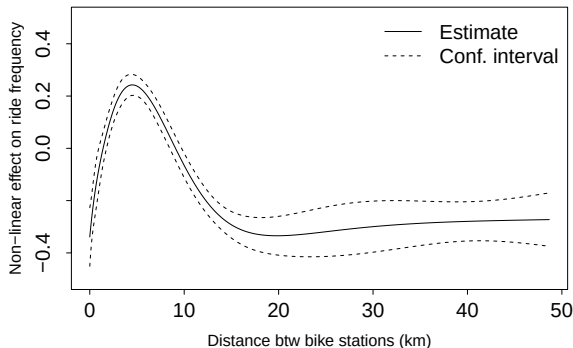
```
gam(y ~ -1+ s(cbind(dist1, dist2),by = cbind(ev.i, nonev.i),
              family= "binomial", data = data)
```

NLE: contribution of distance btw two bike stations



► ... to **sampled likelihood function** ($m=1$)

$$\mathcal{L}^S(\theta) \stackrel{m=1}{=} \prod_{i=1}^n \frac{e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}}{1 + e^{\dots + \left[\sum_{j=1}^q \theta_j [b_j(x_{s_i r_i}^{(3)}(t_i)) - b_j(x_{s_i^* r_i^*}^{(3)}(t_i))] \right] + \dots}} \quad (+ \text{ smoothing penalty})$$



Modeling Random Effects



- ▶ Effect of $v_{sr}(t) = \mathbb{1}\{\text{start station} = s\}$ as $\rightarrow$ **0 - dim spline**

$$\gamma \cdot v_{sr}(t) \approx \sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s' = s\} \quad \gamma = (\gamma_1, \dots, \gamma_{|V|})$$

- ▶ Consider a model with Gaussian random effects, $\gamma \sim \mathcal{N}(0, \sigma^2 I_{|V|})$;
- ▶ Same as fitting a ridge regression model (quadratic penalty term on γ)

Estimating $\gamma \iff$ Estimating $\gamma_{s'}$ -s (and σ^2)

RE: contribution of bike station's popularity as start

► ... to **log-hazard of an interaction**

$$f_{sr} = \dots + \underbrace{\sum_{s' \in V} \gamma_{s'} \mathbb{1}\{s = s'\}}_{\text{random effect}} + \dots$$

► ... to **sampled log-likelihood function (m=1)**

$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left(\frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$

RE: contribution of bike station's popularity as start



► ... to sampled log-likelihood function (m=1)

$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left(\frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$

```
> head(data)
```

```
      y      time      s1      s2 ev.i nonev.i
[1,] 1 0.003427363 31324 31023     1      -1
[2,] 1 0.041364621 31324 31262     1      -1
[3,] 1 0.042194513 31324 31272     1      -1
[4,] 1 0.049880008 31324 31262     1      -1
[5,] 1 0.051420690 31324 31071     1      -1
[6,] 1 0.051820703 31324 31419     1      -1
```

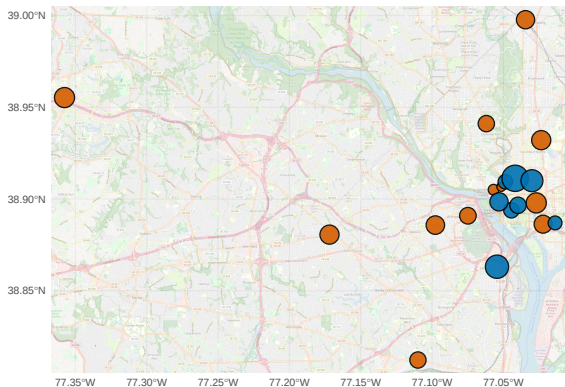
```
stat_pop <- matrix(factor(c(data$s1, data$s2)), ncol = 2)
```

```
gam(y ~ -1+ s(stat_pop, by = cbind(ev.i, nonev.i), bs = "re"),
     family="binomial", data = data)
```

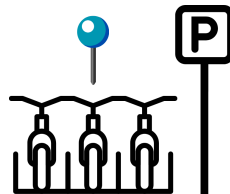
RE: contribution of bike station's popularity as start

► ... to sampled log-likelihood function ($m=1$)

$$\ell^S(\gamma) = \log \mathcal{L}^S(\gamma) \stackrel{m=1}{=} \sum_{i=1}^n \log \left(\frac{e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}}{1 + e^{\dots + \gamma_{s_i} - \gamma_{s_i^*} + \dots}} \right) - \frac{1}{2\sigma_{\text{station}}^2} \gamma \cdot \gamma$$



Gamma Sign
 ○ Negative
 ● Positive





Hands-on

$\approx$ 45 minutes

- ▶ Load the data in R Studio.
- ▶ Create the case-control dataset and compute covariates.
- ▶ Fit models with linear, time-varying, non-linear, and random effects.

Wrapping up



- ▶ (Some) Dynamic networks can be analyzed as temporal sequence of **relational events** between senders and receivers.
- ▶ REMs can capture complex temporal dynamics of interactions through:
 - both **exogenous** and **endogenous** covariates
 - flexible specification of effects (**non-linear, time-varying** and **random**, for now...) easily estimated using efficient **logistic likelihood inference**.



References I

- Federica Bianchi, Edoardo Filippi-Mazzola, Alessandro Lomi, and Ernst C. Wit. Relational event modeling. *Annual Review of Statistics and Its Application*, 11, 2024.
- Martina Boschi, Rūta Juozaitienė, and Ernst C Wit. Mixed additive modelling of global alien species co-invasions of plants and insects. *Journal of the Royal Statistical Society Series C: Applied Statistics*, page qlaf034, 2025.
- Carter T. Butts. A relational event framework for social action. *Sociological Methodology*, 38(1):155–200, 2008.
- David R Cox. Partial likelihood. *Biometrika*, 62(2):269–276, 1975.
- Edoardo Filippi-Mazzola and Ernst C Wit. A stochastic gradient relational event additive model for modelling us patent citations from 1976 to 2022. *Journal of the royal statistical society series C: applied statistics*, 73(4):1008–1024, 2024.



References II

- Jürgen Lerner and Alessandro Lomi. Reliability of relational event model estimates under sampling: How to fit a relational event model to 360 million dyadic events. *Network Science*, 8(1):97–135, 2020.
- Patrick O. Perry and Patrick J. Wolfe. Point process modelling for directed interaction networks. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 75(5):821–849, 2013.
- Christoph Stadtfeld, James Hollway, and Per Block. Dynamic network actor models: Investigating coordination ties through time. *Sociological Methodology*, 47(1):1–40, 2017.
- Duy Vu, Philippa Pattison, and Garry Robins. Relational event models for social learning in moocs. *Social Networks*, 43:121–135, 2015.



Smooth Relational Event Model

To avoid over-fitting, we consider a *penalized partial log-likelihood*;

$$\ell^\nu(\boldsymbol{\theta}) = \ell(\boldsymbol{\theta}) - P_\nu(\boldsymbol{\theta}) = \log(\mathcal{L}(\boldsymbol{\theta})) - \frac{1}{2} \sum_{l=1}^L \nu_l P_l, \quad P_l = \boldsymbol{\theta}_{j_l}^\top S_l \boldsymbol{\theta}_{j_l}, \quad j_l \subseteq \{1, \dots, |\boldsymbol{\theta}|\}$$

- ▶ **Linear:** $S_l = 0$;
- ▶ **Time-Varying / Non-Linear:**
non-sparse S_l ;
- ▶ **Random:**
 - **Penalized quasi-likelihood** for Gaussian random effects:

$$\ell^\nu(\boldsymbol{\beta}, \boldsymbol{\theta}) \approx \ell(\boldsymbol{\beta}, \boldsymbol{\theta}) - \frac{1}{2} \boldsymbol{\theta}^\top \Sigma(\boldsymbol{\psi})^{-1} \boldsymbol{\theta}$$
 (Breslow and Clayton, 1993)
 - S_l is an identity matrix;

